

| Question |     |       | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Marks available |        |     |       |       |      |
|----------|-----|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|--------|-----|-------|-------|------|
|          |     |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | AO1             | AO2    | AO3 | Total | Maths | Prac |
| 6        | (a) | (i)   | Electrons move at right angles to $B$ -field <b>OR</b> LHR mentioned <b>OR</b> accept $Bqv$ or $BIl$ force (anything to do with magnetic force acting on electrons) (1)<br>Force towards back face or shown clearly in diagram (1)<br>(-ve) charge build (up on back face) (1)<br>Electric field hence voltage (ignore “inducing” = neutral)<br><b>OR</b> accept equilibrium when mag force = electric force<br><b>OR</b> accept charge build-up <u>clearly</u> linked to Hall voltage (1) |                 | 4      |     | 4     |       | 4    |
|          |     | (ii)  | $Ee = Bqv$ or similar seen e.g. $Eq = BIl$ or conservation of energy stated (1)<br>$E = \frac{V}{d}$ substituted <b>OR</b> $E = \frac{V}{d}$ and $q = It$ and $v = \frac{l}{t}$ substituted<br>or $Vq = Bqvd$ i.e. conservation of energy applied (1)<br>Algebra seen to give correct answer (1)                                                                                                                                                                                           | 1               | 1<br>1 |     | 3     | 2     |      |
|          |     | (iii) | Any 3 numbers substituted to obtain the 4 <sup>th</sup> (1)<br>Correct answer 4 <sup>th</sup> number & conclusion seen (1)<br>( $7.26 \times 10^{-5} \text{ m s}^{-1}$ is correct drift velocity) 5.4 mm is calculated $d$ , 65.5 nV is calculated Hall pd, 0.187 T is calculated $B$ -field)                                                                                                                                                                                              |                 |        | 2   | 2     | 2     | 2    |
|          |     | (iv)  | Rearrangement of $I = nAve$ ( $n = \frac{I}{Ave}$ ) <b>OR</b> $V_H = \frac{BI}{net}$ known (1)<br>Correct area method i.e. $5.2 \times 0.4 (\times 10^{-6})$ <b>OR</b> $n = \frac{BI}{V_H et}$ (1)<br>Correct answer ( $3.28 \times 10^{28} \text{ m}^{-3}$ ) <b>OR</b> $3.4 \times 10^{28}$ if they use $0.07 \text{ mm s}^{-1}$ (beware <b>ecf</b> may be available for a candidate that uses a previously <b>penalised</b> value instead of the given values)(1)                        |                 | 3      |     | 3     | 3     | 3    |

| Question |     |  | Marking details                                                                                                                                                                                                                                        | Marks available |          |          |           |          |           |
|----------|-----|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|-----------|----------|-----------|
|          |     |  |                                                                                                                                                                                                                                                        | AO1             | AO2      | AO3      | Total     | Maths    | Prac      |
|          | (b) |  | 1 <sup>st</sup> mark - could be either <u>gold or silver</u> (both required for the mark)<br>2 <sup>nd</sup> mark either:<br>Inside range <b>or</b> reference to uncertainty <b>or</b> range is 5.66 – 6.04<br><b>or</b> check the colour of the metal |                 |          | 2        | 2         |          | 2         |
|          |     |  | <b>Question 6 total</b>                                                                                                                                                                                                                                | <b>1</b>        | <b>9</b> | <b>4</b> | <b>14</b> | <b>7</b> | <b>11</b> |